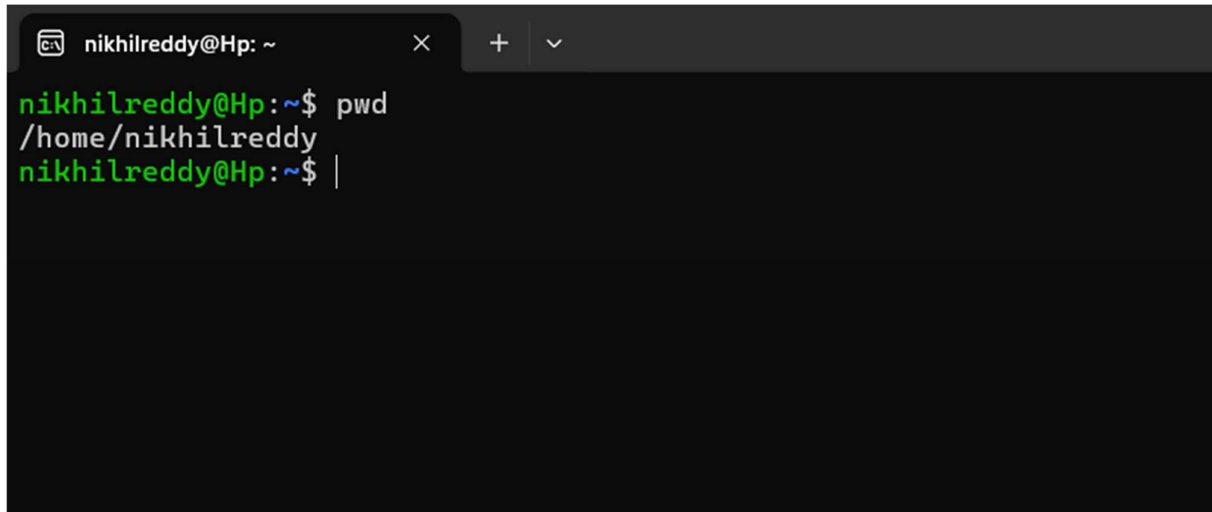


## Topic 1: Basic Linux Commands

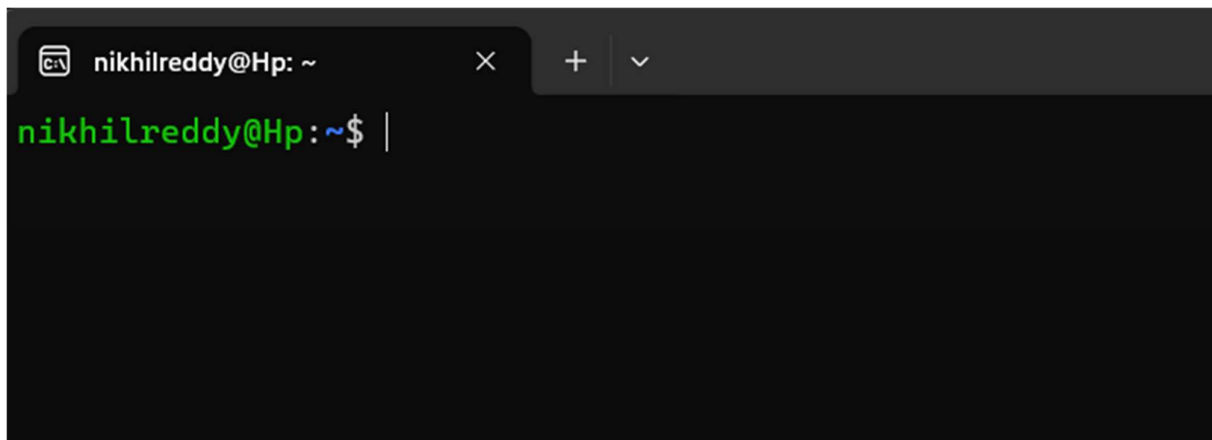
### 1. Display current directory



```
nikhilreddy@Hp: ~  
nikhilreddy@Hp:~$ pwd  
/home/nikhilreddy  
nikhilreddy@Hp:~$ |
```

A terminal window with a dark background. The title bar shows 'nikhilreddy@Hp: ~'. The prompt is 'nikhilreddy@Hp:~\$'. The user enters 'pwd' and the output is '/home/nikhilreddy'. The prompt returns to 'nikhilreddy@Hp:~\$' with a cursor.

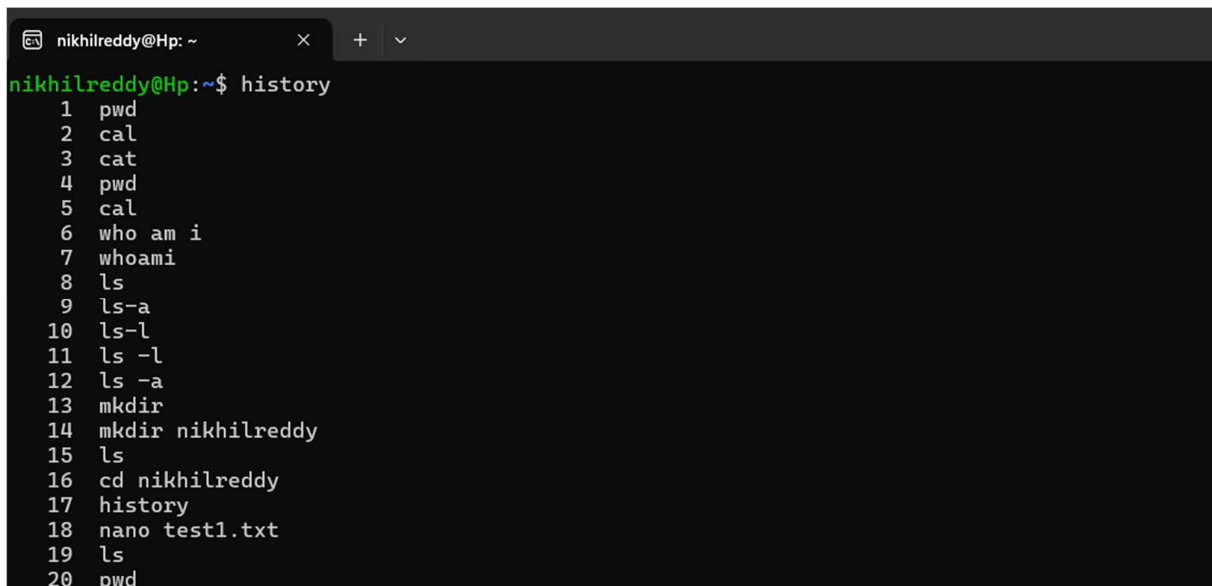
### 2. Clear the terminal



```
nikhilreddy@Hp: ~  
nikhilreddy@Hp:~$ |
```

A terminal window with a dark background. The title bar shows 'nikhilreddy@Hp: ~'. The prompt is 'nikhilreddy@Hp:~\$' with a cursor. The terminal is cleared, showing no previous output.

### 3. Display previously executed commands



```
nikhilreddy@Hp: ~  
nikhilreddy@Hp:~$ history  
1  pwd  
2  cal  
3  cat  
4  pwd  
5  cal  
6  who am i  
7  whoami  
8  ls  
9  ls-a  
10 ls-l  
11 ls -l  
12 ls -a  
13 mkdir  
14 mkdir nikhilreddy  
15 ls  
16 cd nikhilreddy  
17 history  
18 nano test1.txt  
19 ls  
20 pwd
```

A terminal window with a dark background. The title bar shows 'nikhilreddy@Hp: ~'. The prompt is 'nikhilreddy@Hp:~\$'. The user enters 'history' and the output is a list of 20 previously executed commands, numbered 1 to 20. The prompt returns to 'nikhilreddy@Hp:~\$'.

4. Display current username

```
nikhilreddy@Hp:~$ whoami
nikhilreddy
nikhilreddy@Hp:~$ |
```

5. Display date and time

```
nikhilreddy@Hp:~$ date
Tue Aug 11 13:47:33 UTC 2026
nikhilreddy@Hp:~$ |
```

6. Display calendar

```
nikhilreddy@Hp:~$ cal
      August 2026
Su Mo Tu We Th Fr Sa
                1
 2  3  4  5  6  7  8
 9 10 11 12 13 14 15
16 17 18 19 20 21 22
23 24 25 26 27 28 29
30 31
nikhilreddy@Hp:~$ |
```

7. Display system hostname

```
nikhilreddy@Hp:~$ hostname
Hp
nikhilreddy@Hp:~$ |
```

8. Display operating system information

```
nikhilreddy@Hp:~$ uname -a
Linux Hp 6.18.33.2-microsoft-standard-WSL2 #1 SMP PREEMPT_DYNAMIC Thu Jun 18 21:54:43 UTC 2026 x86_64 GNU/Linux
nikhilreddy@Hp:~$ |
```

9. Exit the terminal

```
exitnikhilreddy@Hp:~$ exit|
```

## Topic 2: Directory Management

10. List files and directories

```
nikhilreddy@Hp:~$ ls
example.txt  filename.c  input.txt  nikhil  nikhil.c  nikhil1  nikhil1.c  nikhilprograms  nikhilreddy  output.txt
nikhilreddy@Hp:~$ |
```

#### 11. Display detailed file information

```
nikhilreddy@Hp:~$ ls -l
total 72
-rw-r--r-- 1 nikhilreddy nikhilreddy  0 Jul 31 04:02 example.txt
-rw-r--r-- 1 nikhilreddy nikhilreddy  78 Jul 31 09:36 filename.c
-rw-r--r-- 1 nikhilreddy nikhilreddy 534 Aug 11 13:31 input.txt
-rwxr-xr-x 1 nikhilreddy nikhilreddy 16168 Aug 11 13:31 nikhil
-rw-r--r-- 1 nikhilreddy nikhilreddy  566 Aug 11 03:59 nikhil.c
-rwxr-xr-x 1 nikhilreddy nikhilreddy 16168 Aug 11 13:20 nikhil1
-rw-r--r-- 1 nikhilreddy nikhilreddy  534 Aug 11 13:22 nikhil1.c
-rwxr-xr-x 1 nikhilreddy nikhilreddy 15952 Jul 31 09:36 nikhilprograms
drwxr-xr-x 2 nikhilreddy nikhilreddy  4096 Jul 28 04:07 nikhilreddy
-rw-r--r-- 1 nikhilreddy nikhilreddy  534 Aug 11 13:31 output.txt
nikhilreddy@Hp:~$ |
```

#### 12. Display all files, including hidden files

```
nikhilreddy@Hp:~$ ls -a
.      .bash_history  .bashrc  .config  .motd_shown  example.txt  input.txt  nikhil.c  nikhil1.c  nikhilreddy
..     .bash_logout  .cache   .local   .profile     filename.c   nikhil     nikhil1   nikhilprograms  output.txt
nikhilreddy@Hp:~$ |
```

#### 13. Change the current directory

```
nikhilreddy@Hp:~$ cd Documents
-bash: cd: Documents: No such file or directory
nikhilreddy@Hp:~$ |
```

#### 14. Move to the parent directory

```
nikhilreddy@Hp:~$ cd ..
nikhilreddy@Hp:/home$ |
```

#### 15. Create a new directory

```
nikhilreddy@Hp:/home$ mkdir OSLab
mkdir: Permission denied
nikhilreddy@Hp:/home$ |
```

#### 16. Remove an empty directory

```
nikhilreddy@Hp:/home$ rmdir Test
rmdir: failed to remove 'Test': No such file or directory
nikhilreddy@Hp:/home$ |
```

#### 17. Search for C files

```
nikhilreddy@Hp:/home$ find . -name "*.c"
./nikhilreddy/nikhil1.c
./nikhilreddy/filename.c
./nikhilreddy/nikhil.c
nikhilreddy@Hp:/home$ |
```

18. Quickly locate a file

```
nikhilreddy@Hp:/home$ locate hello.c
Command 'locate' not found, but can be installed with:
sudo apt install plocate
nikhilreddy@Hp:/home$ |
```

### Topic 3: File Management

19. Create a new empty file

```
nikhilreddy@Hp:/home$ touch file1.txt
touch: cannot touch 'file1.txt': Permission denied
nikhilreddy@Hp:/home$ |
```

20. Display file contents

```
nikhilreddy@Hp:/home$ cat file1.txt
cat: file1.txt: No such file or directory
nikhilreddy@Hp:/home$ |
```

21. Copy a file

```
nikhilreddy@Hp:/home$ cp a.txt b.txt
cp: cannot stat 'a.txt': No such file or directory
nikhilreddy@Hp:/home$ |
```

22. Move or rename a file

```
nikhilreddy@Hp:/home$ locate hello.c
Command 'locate' not found, but can be installed with:
sudo apt install plocate
nikhilreddy@Hp:/home$ touch file1.txt
touch: cannot touch 'file1.txt': Permission denied
nikhilreddy@Hp:/home$ cat file1.txt
cat: file1.txt: No such file or directory
nikhilreddy@Hp:/home$ cp a.txt b.txt
cp: cannot stat 'a.txt': No such file or directory
nikhilreddy@Hp:/home$ mv old.txt new.txt
mv: cannot stat 'old.txt': No such file or directory
nikhilreddy@Hp:/home$ |
```

23. Delete a file

```

nikhilreddy@Hp:~$ ls -a
.  .bash_history  .bashrc  .config  .motd_shown  example.txt  input.txt  nikhil.
.. .bash_logout  .cache   .local   .profile     filename.c   nikhil    nikhil1
nikhilreddy@Hp:~$ cd Documents
-bash: cd: Documents: No such file or directory
nikhilreddy@Hp:~$ cd ..
nikhilreddy@Hp:/home$ mkdir OSLab
mkdir: Permission denied
nikhilreddy@Hp:/home$ rmdir Test
rmdir: failed to remove 'Test': No such file or directory
nikhilreddy@Hp:/home$ find . -name "*.c"
./nikhilreddy/nikhil1.c
./nikhilreddy/filename.c
./nikhilreddy/nikhil.c
nikhilreddy@Hp:/home$ Quickly locate a file
Quickly: command not found
nikhilreddy@Hp:/home$ locate hello.c
Command 'locate' not found, but can be installed with:
sudo apt install plocate
nikhilreddy@Hp:/home$ touch file1.txt
touch: cannot touch 'file1.txt': Permission denied
nikhilreddy@Hp:/home$ cat file1.txt
cat: file1.txt: No such file or directory
nikhilreddy@Hp:/home$ cp a.txt b.txt
cp: cannot stat 'a.txt': No such file or directory
nikhilreddy@Hp:/home$ mv old.txt new.txt
mv: cannot stat 'old.txt': No such file or directory
nikhilreddy@Hp:/home$ rm test.txt
rm: cannot remove 'test.txt': No such file or directory
nikhilreddy@Hp:/home$ |

```

24. Display the first lines of a file

```

nikhilreddy@Hp:/home$ cat file1.txt
cat: file1.txt: No such file or directory
nikhilreddy@Hp:/home$ cp a.txt b.txt
cp: cannot stat 'a.txt': No such file or directory
nikhilreddy@Hp:/home$ mv old.txt new.txt
mv: cannot stat 'old.txt': No such file or directory
nikhilreddy@Hp:/home$ rm test.txt
rm: cannot remove 'test.txt': No such file or directory
nikhilreddy@Hp:/home$ head hello.c
head: cannot open 'hello.c' for reading: No such file or directory
nikhilreddy@Hp:/home$ |

```

25. Display the last lines of a file

```

head: cannot open 'hello.c' for reading: No such file or directory
nikhilreddy@Hp:/home$ tail hello.c
tail: cannot open 'hello.c' for reading: No such file or directory
nikhilreddy@Hp:/home$ |

```